

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching agents to autonomously decide when to act directly, self-reflect, or consult external knowledge during planning.

arxiv.org/abs/2504.03553

Technical Presentation • ML Research

The Problem: Indiscriminate Knowledge Injection

Trajectory Imitation

e.g., ReAct

Blindly mimics gold trajectories without assessing situational need

Trial-and-Error

e.g., Reflexion

Refines via external feedback at every step regardless of necessity

Knowledge-Augmented

e.g., KnowAgent / WKM

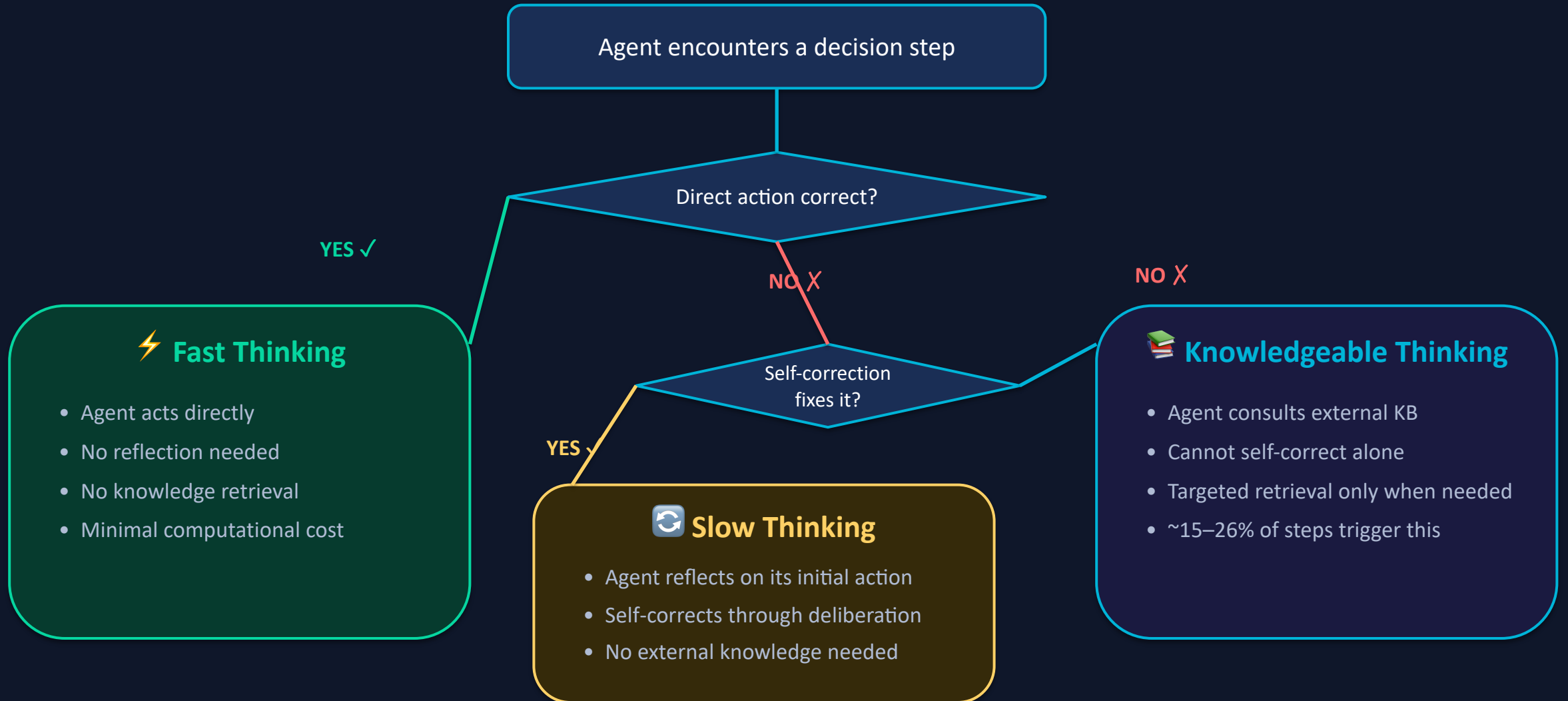
Injects domain knowledge at 100% of steps even when agent already knows

▼ All use a "flood irrigation" approach

- Fragile planning — patterns collapse on unexpected inputs
- Unnecessarily high inference costs — 100% knowledge injection at every step
- Humans possess situational self-awareness; current agents do not

KnowSelf brings metacognitive self-awareness to LLM agents — knowing when to think, reflect, or seek help.

Three Situational Thinking Modes



KnowSelf Framework: Data → Training → Inference

Step 1

Self-Awareness Data Construction

Agent self-explores environments
Heuristic criterion classifies each step:
Predicted = Gold → Fast token
Reflection fixes → Slow token
Otherwise → Knowledge token
Special tokens inserted into trajectories



Step 2

Two-Stage Training

Stage A: Supervised Fine-Tuning (SFT)
Teaches initial self-awareness patterns
from labeled trajectory data
Stage B: Reward-based Policy Optim. (RPO)
Refines special token generation
via reward signal alignment



Step 3

Self-Aware Inference

Agent generates special tokens at runtime
Token triggers appropriate action:
Fast → direct action execution
Slow → internal reflection loop
Knowledge → external KB retrieval
Selective, efficient knowledge use

Lightweight pipeline: heuristic labeling on self-explored trajectories — no expensive human annotation required.

Qualitative Examples: KnowSelf vs. Strong Baselines

Task: "Put a hot mug in cabinet"

✗ OpenAI-O1

Opens microwave, then incorrectly tries to take an egg from it instead of heating the mug

✓ KnowSelf (Llama-8B)

Uses Reflection token to realize it should heat the mug using the already-open microwave

Mode:  Slow Thinking

Self-correction through reflection was sufficient

Task: "Put a saltshaker in drawer"

✗ DeepSeek-R1

Reflects that the drawer is open, tries to close it first (which fails)

✓ KnowSelf (Llama-8B)

Applies Knowledge token: agent should place the object without removing unrelated items

Mode:  Knowledgeable Thinking

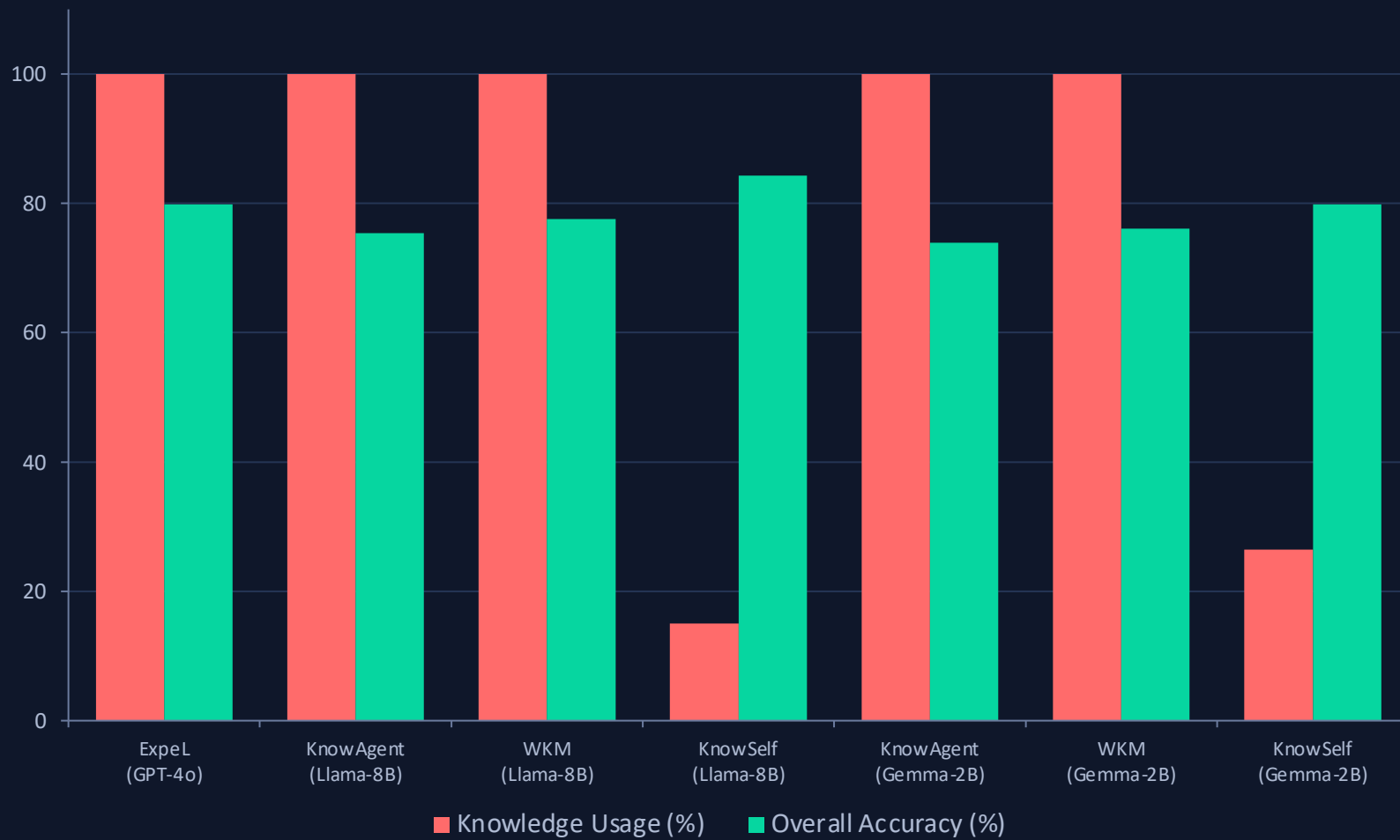
External knowledge was necessary for correct action

Main Results: ALFWorld Success Rates (%)

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

KnowSelf (Llama-8B) 84.33% surpasses GPT-4o Reflexion 85.07% with only 15% knowledge — the best efficiency-accuracy tradeoff.

Minimal Knowledge, Maximum Impact: Efficiency Advantage



Key Insights

- KnowSelf (Llama-8B) uses only 15% knowledge but achieves 84.33% — the highest accuracy among all methods.
- Methods using 100% knowledge (KnowAgent, WKM, ExpeL) consistently score lower, confirming that indiscriminate injection hurts.
- Even Gemma-2B with KnowSelf (26% knowledge) matches GPT-4o ExpeL (100% knowledge) at 79.85%.
- Selective knowledge use scales effectively across model sizes — the efficiency gap is model-agnostic.

Key Takeaways & Future Directions

Key Takeaways

- 1 Indiscriminate knowledge injection is suboptimal**
100% knowledge methods (ExpeL, KnowAgent, WKM) consistently underperform selective approaches.
- 2 Agents can learn situational self-awareness**
Special-token mechanism with heuristic labeling — no expensive human annotation required.
- 3 KnowSelf (Llama-8B) outperforms GPT-4o baselines**
84.33% vs. ReAct 76.12% and ExpeL 79.85% while using only 15% knowledge.
- 4 Only 15–26% knowledge usage needed**
Dramatic reduction in inference cost with no accuracy penalty — actually improves it.
- 5 Scales across model sizes**
Even Gemma-2B reaches 79.85% matching GPT-4o ExpeL, validating cross-scale generalization.

Future Directions

- Extend to multi-agent collaboration scenarios where situational awareness must coordinate across agents
- Apply to richer environments beyond ALFWorld (WebShop, ScienceWorld, real-world APIs)
- Investigate learned token distributions to understand when models struggle most
- Combine with curriculum learning — gradually reduce knowledge scaffolding as agents improve
- Explore integration with retrieval-augmented generation (RAG) for dynamic knowledge bases